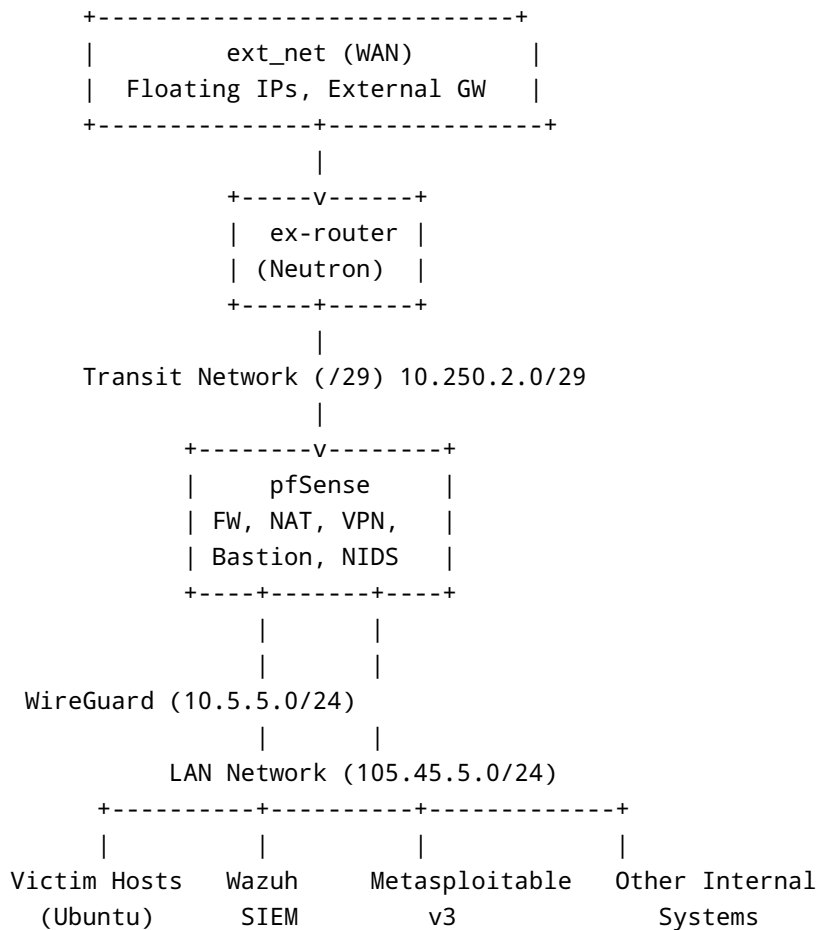


Full Environment Diagram Documentation

Purpose

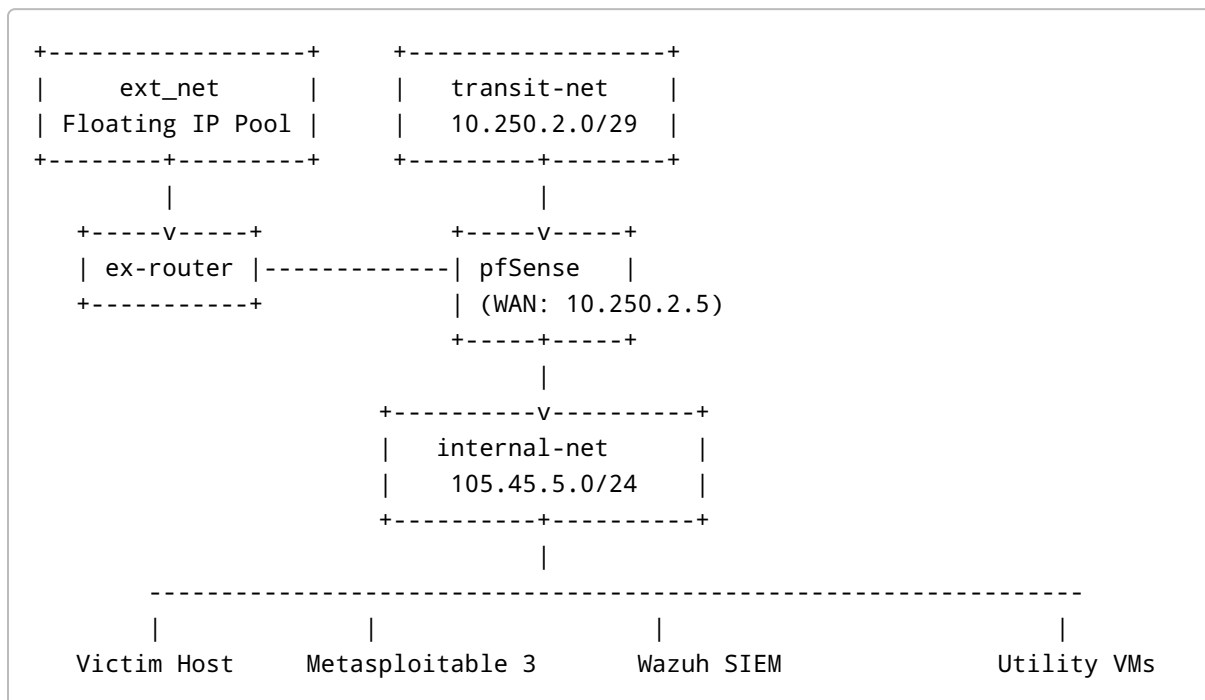
This document provides complete technical documentation of all network, routing, and system diagrams used in the Virtual SOC environment. It defines every architectural component, their relationships, and the traffic flow paths. These diagrams are intended for engineering, instructional, and auditing use.

1. High-Level Architecture Diagram

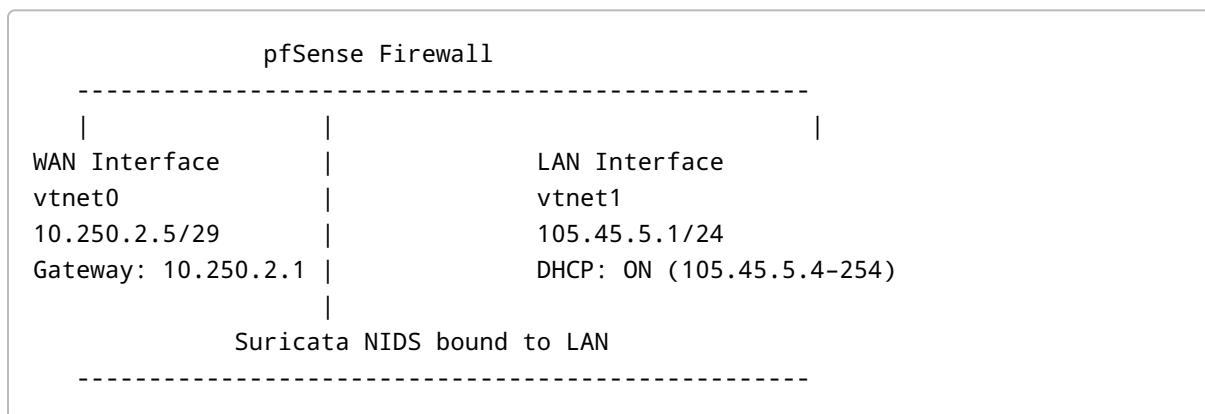


This diagram represents the complete multi-segment SOC: ext_net on top, pfSense in the middle, full internal network at the bottom, and WireGuard access on the side.

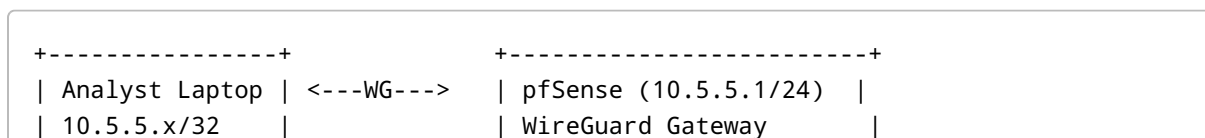
2. OpenStack Logical Network Diagram

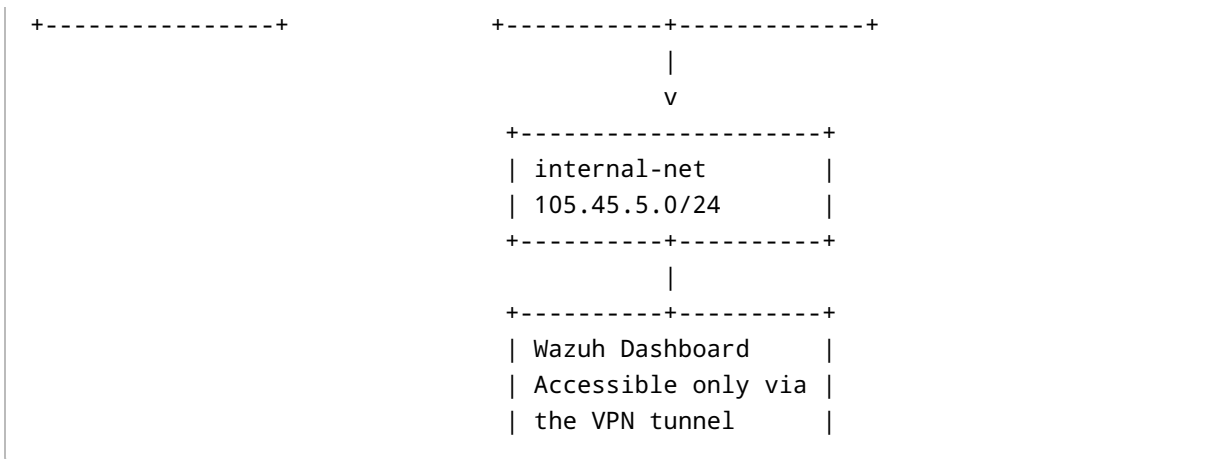


3. pfSense Interface Diagram

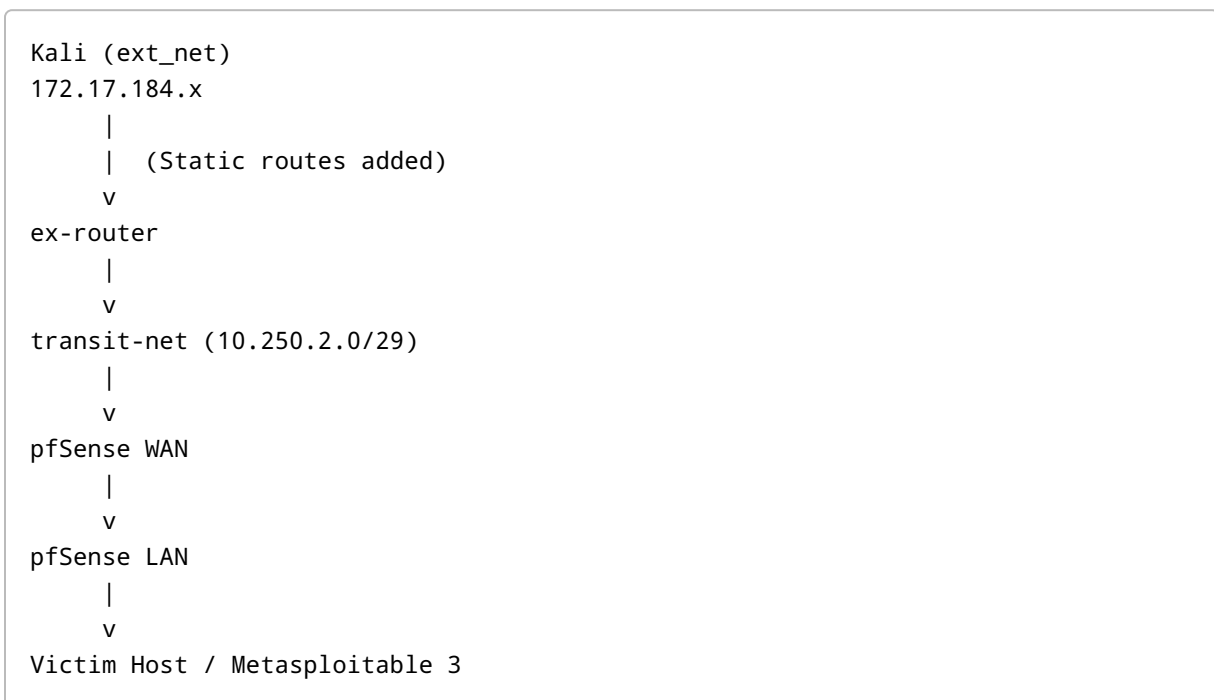


4. WireGuard VPN Diagram



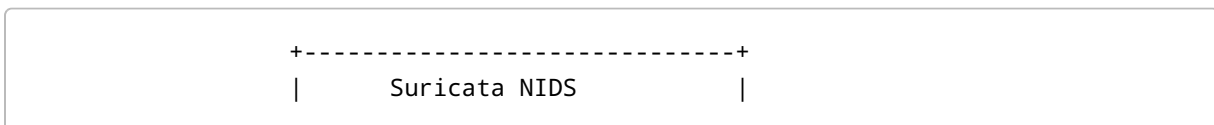


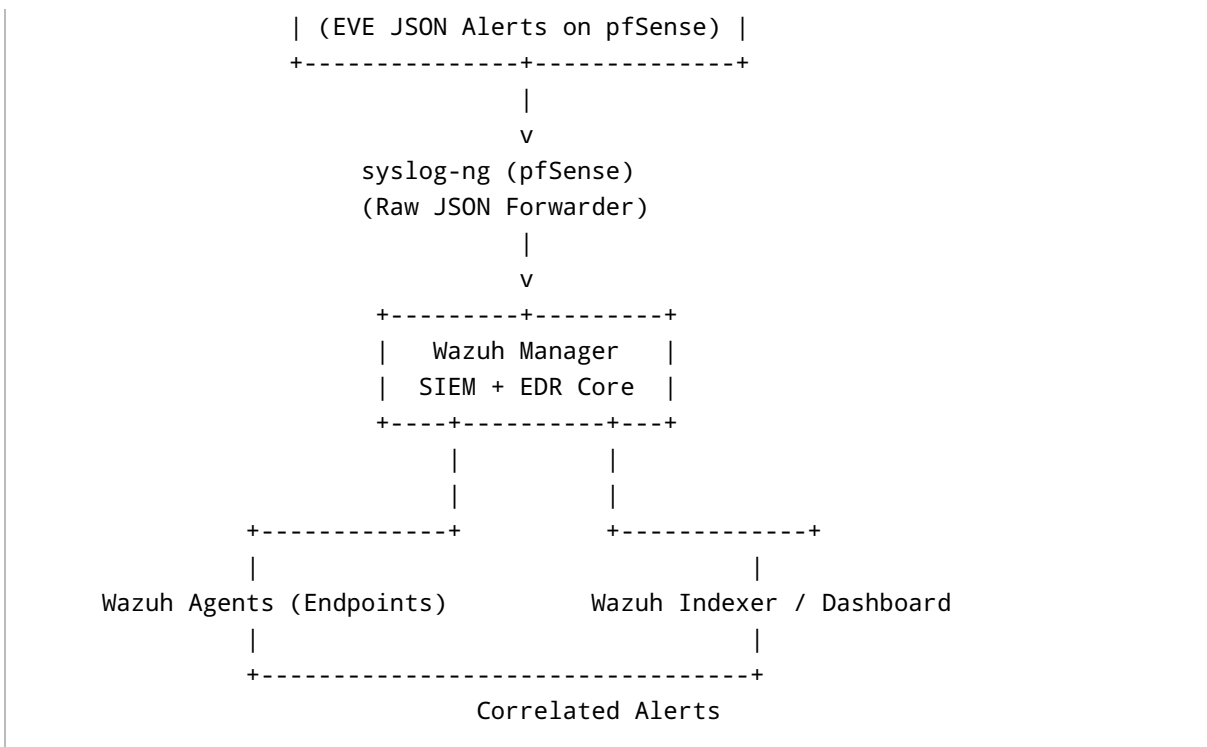
5. Attack Path Diagram (Kali → Victim)



All attack traffic flows through pfSense → Suricata → Wazuh.

6. Telemetry Pipeline Diagram





7. End-to-End SOC Data Flow Summary

1. Attacker traffic arrives from ext_net.
2. Traffic routes through ex-router → transit-net.
3. pfSense receives traffic on WAN.
4. pfSense forwards traffic to LAN.
5. Suricata monitors and logs alerts.
6. syslog-ng forwards alerts to Wazuh.
7. Wazuh correlates:
8. Suricata alerts
9. Endpoint logs (via agents)
10. pfSense system logs
11. Dashboard displays real-time SOC alerts.

8. Purpose of Diagrams

These diagrams: - Provide full clarity on system architecture. - Enable replication and troubleshooting. - Assist in academic instruction and SOC demonstrations. - Serve as authoritative documentation for future expansion.

They collectively represent the entire Virtual SOC ecosystem and its operational data flow.